KB-EWS supplementary material

This file accompanies the short communication KB-EWS. It records the checks that are summarised in the main text and the files needed to rebuild them.

S1. Reproducibility

The program kbews.py, included in this supplement, downloads the public Crunchbase CSV conversion, rebuilds the cohort, revises the rule base, fits the five scores, and writes Table 1, Table 2, Figures 1–3, metrics.json, and warning_cards.csv. The random seed is 7. Raw Crunchbase files are not redistributed. Source: Crunchbase Data Export of 4 December 2015, CSV conversion by Peter Tripp, github.com/notpeter/crunchbase-data, commit be8711fd, licence CC BY-NC 4.0. Commercial use of the extract requires a Crunchbase licence.

S2. Retention-margin sensitivity

The margin is applied to training-year mean rule strength only. The booster is not refit.

Margin 0.00: 4 rules retained (R1, R2, R3, R6); knowledge-base ROC-AUC 0.618; precision at 30% 0.633.

Margin 0.02: 4 rules retained (R1, R2, R3, R6); knowledge-base ROC-AUC 0.618; precision at 30% 0.633.

Margin 0.05: 3 rules retained (R2, R3, R6); knowledge-base ROC-AUC 0.608; precision at 30% 0.638.

S3. Standardized logistic coefficients on the training years

Positive coefficients raise the log-odds of later closure. Predictors are standardized on 2006–2010. Reference levels are a venture round and the United States.

log_first_usd: -0.511

round_type_angel: 0.333

log1p_ecosystem: -0.328

log1p_days: -0.276

country_missing: 0.208

round_type_seed: 0.163

n_categories: -0.154

digital_depth: 0.132

country_CHN: -0.099

country_CAN: -0.086

country_FRA: 0.073

country_DEU: 0.067

country_OTHER: -0.048

country_GBR: 0.028

country_ISR: 0.011

round_type_other: -0.000

S4. Precision across review budgets

These are the point estimates drawn in Figure 2. The budget was not chosen on the test labels.

Knowledge base: 10%=0.635, 15%=0.649, 20%=0.646, 25%=0.630, 30%=0.633, 40%=0.603, 50%=0.592

Logistic regression: 10%=0.892, 15%=0.856, 20%=0.816, 25%=0.788, 30%=0.751, 40%=0.712, 50%=0.696

Decision tree: 10%=0.797, 15%=0.784, 20%=0.769, 25%=0.766, 30%=0.751, 40%=0.729, 50%=0.704

Gradient boosting: 10%=0.865, 15%=0.820, 20%=0.810, 25%=0.799, 30%=0.760, 40%=0.739, 50%=0.688

Stacked fusion: 10%=0.865, 15%=0.829, 20%=0.789, 25%=0.799, 30%=0.769, 40%=0.732, 50%=0.688

S5. Held-out rule firing

R1 Resource strain (retained, CF +0.55): strength at least 0.5 for 0.157 of later closures and 0.097 of recorded exits.

R2 Unfunded scope (retained, CF +0.45): strength at least 0.5 for 0.479 of later closures and 0.299 of recorded exits.

R3 Shallow digital base (retained, CF +0.35): strength at least 0.5 for 0.465 of later closures and 0.258 of recorded exits.

R4 Isolated entry (withdrawn, CF +0.30): strength at least 0.5 for 0.098 of later closures and 0.102 of recorded exits.

R5 Unfocused entry (withdrawn, CF +0.25): strength at least 0.5 for 0.463 of later closures and 0.482 of recorded exits.

R6 Resourced focus (retained, CF -0.50): strength at least 0.5 for 0.085 of later closures and 0.188 of recorded exits.

R7 Embedded start (withdrawn, CF -0.30): strength at least 0.5 for 0.479 of later closures and 0.504 of recorded exits.